

NT-C-033 | Ethical Hacking | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Cybersecurity
Target learners	Learners seeking authorised security-testing foundations.
Prerequisites	Networking, Linux and cybersecurity basics; all work must use authorised labs.
Approved duration	14-18 weeks / 160-220 guided hours.
Final outcome	Plan and document an authorised security assessment, validate findings safely and recommend remediation.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Isolated authorised lab, Linux security distribution, network/web testing tools, proxy, scanner and reporting templates. No production testing without written approval.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-033 | Ethical Hacking | NT-C-033-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Lawful authorisation, rules of engagement and lab safety; Passive information gathering and asset mapping
Completion evidence	Signed scope exercise; Recon report
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Ethics and scope

Objective: Lawful authorisation, rules of engagement and lab safety

Observable evidence: Signed scope exercise

Concept	Approved knowledge statement	Application behaviour	Evidence
Lawful authorisation	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when lawful authorisation is appropriate; apply it in a Ethical Hacking task; verify the result.	Signed scope exercise
rules of engagement	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when rules of engagement is appropriate; apply it in a Ethical Hacking task; verify the result.	Signed scope exercise
lab safety	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when lab safety is appropriate; apply it in a Ethical Hacking task; verify the result.	Signed scope exercise

Module 2 - Reconnaissance

Objective: Passive information gathering and asset mapping

Observable evidence: Recon report

Concept	Approved knowledge statement	Application behaviour	Evidence
Passive information gathering	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when passive information gathering is appropriate; apply it in a Ethical Hacking task; verify the result.	Recon report
asset mapping	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when asset mapping is appropriate; apply it in a Ethical Hacking task; verify the result.	Recon report

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Signed scope exercise; Recon report, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-033 | Ethical Hacking | NT-C-033-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Ports, services, enumeration and validation; OWASP-oriented input, session and access-control checks
Completion evidence	Asset/service inventory; Web assessment notes
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Scanning

Objective: Ports, services, enumeration and validation

Observable evidence: Asset/service inventory

Concept	Approved knowledge statement	Application behaviour	Evidence
Ports	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ports is appropriate; apply it in a Ethical Hacking task; verify the result.	Asset/service inventory
services	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when services is appropriate; apply it in a Ethical Hacking task; verify the result.	Asset/service inventory
enumeration	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when enumeration is appropriate; apply it in a Ethical Hacking task; verify the result.	Asset/service inventory
validation	Checking input or output against defined rules before trusting it. In Ethical Hacking, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a Ethical Hacking task; verify the result.	Asset/service inventory

Module 4 - Web testing

Objective: OWASP-oriented input, session and access-control checks

Observable evidence: Web assessment notes

Concept	Approved knowledge statement	Application behaviour	Evidence
OWASP-oriented input	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when owasp-oriented input is appropriate; apply it in a Ethical Hacking task; verify the result.	Web assessment notes
session	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when session is appropriate; apply it in a Ethical Hacking task; verify the result.	Web assessment notes
access-control checks	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when access-control checks is appropriate; apply it in a Ethical Hacking task; verify the result.	Web assessment notes

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Asset/service inventory; Web assessment notes, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-033 | Ethical Hacking | NT-C-033-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Configuration, patch and credential-risk assessment; Segmentation, wireless concepts and traffic review
Completion evidence	Host findings; Network findings
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - System testing

Objective: Configuration, patch and credential-risk assessment

Observable evidence: Host findings

Concept	Approved knowledge statement	Application behaviour	Evidence
Configuration	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when configuration is appropriate; apply it in a Ethical Hacking task; verify the result.	Host findings
patch	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when patch is appropriate; apply it in a Ethical Hacking task; verify the result.	Host findings
credential-risk assessment	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when credential-risk assessment is appropriate; apply it in a Ethical Hacking task; verify the result.	Host findings

Module 6 - Network testing

Objective: Segmentation, wireless concepts and traffic review

Observable evidence: Network findings

Concept	Approved knowledge statement	Application behaviour	Evidence
Segmentation	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when segmentation is appropriate; apply it in a Ethical Hacking task; verify the result.	Network findings
wireless concepts	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when wireless concepts is appropriate; apply it in a Ethical Hacking task; verify the result.	Network findings
traffic review	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when traffic review is appropriate; apply it in a Ethical Hacking task; verify the result.	Network findings

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Host findings; Network findings, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-033 | Ethical Hacking | NT-C-033-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Controlled proof, evidence, severity and fixes; Executive/technical report, retest and closure
Completion evidence	Finding records; Authorised assessment report
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Validation and remediation

Objective: Controlled proof, evidence, severity and fixes

Observable evidence: Finding records

Concept	Approved knowledge statement	Application behaviour	Evidence
Controlled proof	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when controlled proof is appropriate; apply it in a Ethical Hacking task; verify the result.	Finding records
evidence	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when evidence is appropriate; apply it in a Ethical Hacking task; verify the result.	Finding records
severity	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when severity is appropriate; apply it in a Ethical Hacking task; verify the result.	Finding records
fixes	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when fixes is appropriate; apply it in a Ethical Hacking task; verify the result.	Finding records

Module 8 - Reporting capstone

Objective: Executive/technical report, retest and closure

Observable evidence: Authorised assessment report

Concept	Approved knowledge statement	Application behaviour	Evidence
Executive/technical report	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when executive/technical report is appropriate; apply it in a Ethical Hacking task; verify the result.	Authorised assessment report
retest	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when retest is appropriate; apply it in a Ethical Hacking task; verify the result.	Authorised assessment report
closure	A defined competency within Ethical Hacking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when closure is appropriate; apply it in a Ethical Hacking task; verify the result.	Authorised assessment report

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Finding records; Authorised assessment report, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-033 | Ethical Hacking | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-033-SK01	Ethics and scope	Prerequisite foundation	Signed scope exercise	Critical
NT-C-033-SK02	Reconnaissance	M01 plus earlier foundations	Recon report	Critical
NT-C-033-SK03	Scanning	M02 plus earlier foundations	Asset/service inventory	Supporting
NT-C-033-SK04	Web testing	M03 plus earlier foundations	Web assessment notes	Supporting
NT-C-033-SK05	System testing	M04 plus earlier foundations	Host findings	Supporting
NT-C-033-SK06	Network testing	M05 plus earlier foundations	Network findings	Critical
NT-C-033-SK07	Validation and remediation	M06 plus earlier foundations	Finding records	Supporting
NT-C-033-SK08	Reporting capstone	M07 plus earlier foundations	Authorised assessment report	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-033 | Ethical Hacking | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-033-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Ethics and scope and Reconnaissance?	The learner should provide Signed scope exercise; Recon report and explain how the work applies Lawful authorisation, rules of engagement and lab safety; Passive information gathering and asset mapping. Recall alone is insufficient.
NT-C-033-S01-EX02	Scenario/remediation	A learner reports completing Reconnaissance but cannot produce Recon report. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-033-S01, request Recon report, and allow a new equivalent test only after remediation.
NT-C-033-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Scanning and Web testing?	The learner should provide Asset/service inventory; Web assessment notes and explain how the work applies Ports, services, enumeration and validation; OWASP-oriented input, session and access-control checks. Recall alone is insufficient.
NT-C-033-S02-EX02	Scenario/remediation	A learner reports completing Web testing but cannot produce Web assessment notes. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-033-S02, request Web assessment notes, and allow a new equivalent test only after remediation.
NT-C-033-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of System testing and Network testing?	The learner should provide Host findings; Network findings and explain how the work applies Configuration, patch and credential-risk assessment; Segmentation, wireless concepts and traffic review. Recall alone is insufficient.
NT-C-033-S03-EX02	Scenario/remediation	A learner reports completing Network testing but cannot produce Network findings. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-033-S03, request Network findings, and allow a new equivalent test only after remediation.
NT-C-033-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Validation and remediation and Reporting capstone?	The learner should provide Finding records; Authorised assessment report and explain how the work applies Controlled proof, evidence, severity and fixes; Executive/technical report, retest and closure. Recall alone is insufficient.
NT-C-033-S04-EX02	Scenario/remediation	A learner reports completing Reporting capstone but cannot produce Authorised assessment report. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-033-S04, request Authorised assessment report, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-033 | Ethical Hacking | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Recon and scope pack	Authorisation and asset mapping	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Web lab assessment	Evidence and remediation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	VAPT report	Testing, severity, retest and presentation	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-033 | Ethical Hacking | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	VAPT Trainee	Scope 34% [critical]; testing 33% [critical]; reporting 33%	Authorised lab report	If a critical skill is below 60 or authorised lab report is missing, return Improve First or Needs Evidence with the linked module/project.
2	Junior Penetration Testing Associate	Evidence 34% [critical]; validation 33% [critical]; remediation 33%	Assessment portfolio	If a critical skill is below 60 or assessment portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
3	Security Testing Trainee	Web/network checks 50% [critical]; documentation 50% [critical]	Retest evidence	If a critical skill is below 60 or retest evidence is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-033 | Ethical Hacking | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Ethical Hacking for?	Learners seeking authorised security-testing foundations.
What should I know before starting?	Networking, Linux and cybersecurity basics; all work must use authorised labs.
How long is the approved learning journey?	14-18 weeks / 160-220 guided hours.
What will I be able to demonstrate?	Plan and document an authorised security assessment, validate findings safely and recommend remediation.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Ethical Hacking knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

course_code=NT-C-033; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12